

FUNDAMENTALS OF DATA MINING

Lecture

Association Rule Mining – Apriori Algorithm

1

Association Rule Mining

- Given a set of transactions, find rules that will predict the occurrence of an item based on the occurrences of other items in the transaction

Market-Basket transactions

TID	Items
1	Bread, Milk
2	Bread, Diaper, Beer, Eggs
3	Milk, Diaper, Beer, Coke
4	Bread, Milk, Diaper, Beer
5	Bread, Milk, Diaper, Coke

Example of Association Rules

$\{\text{Diaper}\} \rightarrow \{\text{Beer}\},$
 $\{\text{Milk, Bread}\} \rightarrow \{\text{Eggs, Coke}\},$
 $\{\text{Beer, Bread}\} \rightarrow \{\text{Milk}\},$

Implication means co-occurrence, not causality!

2

Definition: Frequent Itemset

Itemset

- A collection of one or more items
 - Example: {Milk, Bread, Diaper}
- k-itemset
 - An itemset that contains k items

Support count (σ)

- Frequency of occurrence of an itemset
- E.g. $\sigma(\{\text{Milk, Bread, Diaper}\}) = 2$

Support

- Fraction of transactions that contain an itemset
- E.g. $s(\{\text{Milk, Bread, Diaper}\}) = 2/5$

Frequent Itemset

- An itemset whose support is greater than or equal to a *minsup* threshold

TID	Items
1	Bread, Milk
2	Bread, Diaper, Beer, Eggs
3	Milk, Diaper, Beer, Coke
4	Bread, Milk, Diaper, Beer
5	Bread, Milk, Diaper, Coke

3

Definition: Association Rule

Association Rule

- An implication expression of the form $X \rightarrow Y$, where X and Y are itemsets
- Example: $\{\text{Milk, Diaper}\} \rightarrow \{\text{Beer}\}$

Rule Evaluation Metrics

- Support (s)
 - Fraction of transactions that contain both X and Y
- Confidence (c)
 - Measures how often items in Y appear in transactions that contain X

Example:

$\{\text{Milk, Diaper}\} \Rightarrow \text{Beer}$

$$s = \frac{\sigma(\text{Milk, Diaper, Beer})}{|T|} = \frac{2}{5} = 0.4$$

$$c = \frac{\sigma(\text{Milk, Diaper, Beer})}{\sigma(\text{Milk, Diaper})} = \frac{2}{3} = 0.67$$

4

Association Rule Mining Task

- Given a set of transactions T, the goal of association rule mining is to find all rules having

- support $\geq \text{minsup}$ threshold
- confidence $\geq \text{minconf}$ threshold

Brute-force approach:

- List all possible association rules
 - Compute the support and confidence for each rule
 - Prune rules that fail the *minsup* and *minconf* thresholds
- $\Rightarrow$ **Computationally prohibitive!**

5

Mining Association Rules

TID	Items
1	Bread, Milk
2	Bread, Diaper, Beer, Eggs
3	Milk, Diaper, Beer, Coke
4	Bread, Milk, Diaper, Beer
5	Bread, Milk, Diaper, Coke

Example of Rules:

$\{\text{Milk, Diaper}\} \rightarrow \{\text{Beer}\} \ (s=0.4, c=0.67)$
 $\{\text{Milk, Beer}\} \rightarrow \{\text{Diaper}\} \ (s=0.4, c=1.0)$
 $\{\text{Diaper, Beer}\} \rightarrow \{\text{Milk}\} \ (s=0.4, c=0.67)$
 $\{\text{Beer}\} \rightarrow \{\text{Milk, Diaper}\} \ (s=0.4, c=0.67)$
 $\{\text{Diaper}\} \rightarrow \{\text{Milk, Beer}\} \ (s=0.4, c=0.5)$
 $\{\text{Milk}\} \rightarrow \{\text{Diaper, Beer}\} \ (s=0.4, c=0.5)$

Observations:

- All the above rules are binary partitions of the same itemset: {Milk, Diaper, Beer}
- Rules originating from the same itemset have identical support but can have different confidence
- Thus, we may decouple the support and confidence requirements

6

Mining Association Rules

- Two-step approach:
 1. **Frequent Itemset Generation**
 - Generate all itemsets whose support $\geq$ minsup
 2. **Rule Generation**
 - Generate high confidence rules from each frequent itemset, where each rule is a binary partitioning of a frequent itemset
- Frequent itemset generation is still computationally expensive....!!!

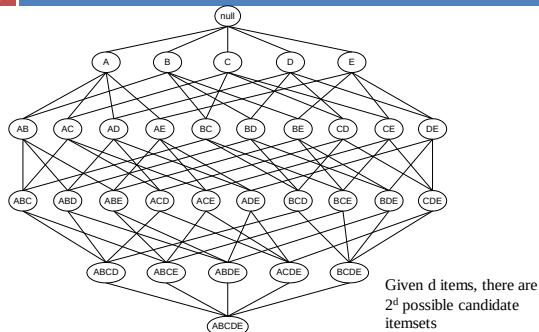
7

Association Rule Mining

- Discovers patterns with frequency above the minimum support threshold. Therefore,
- Association rule mining is not recommended for finding associations involving rare events in problem domains with a large number of items.
- Classification or anomaly detection may be more suitable for discovering rare events when the data has a high number of attributes.

8

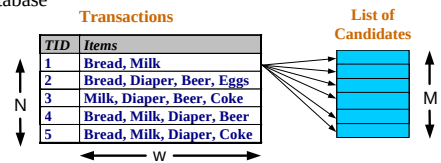
Frequent Itemset Generation



9

Frequent Itemset Generation

- Brute-force approach:
 - Each itemset in the lattice is a **candidate** frequent itemset
 - Count the support of each candidate by scanning the database



10

Frequent Itemset Generation Strategies

- Reduce the **number of candidates** (M)
 - Complete search: $M=2^d$
 - Use pruning techniques to reduce M
- Reduce the **number of transactions** (N)
 - Reduce size of N as the size of itemset increases
 - Used by DHP and vertical-based mining algorithms
- Reduce the **number of comparisons** (NM)
 - Use efficient data structures to store the candidates or transactions
 - No need to match every candidate against every transaction

11

Reducing Number of Candidates

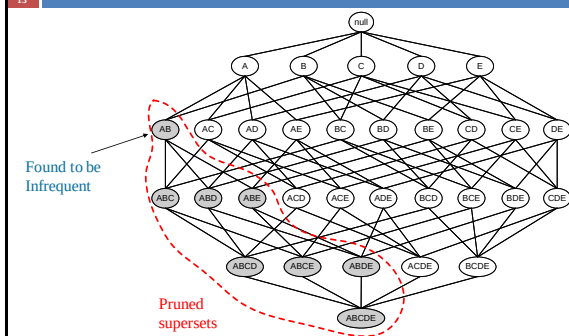
- **Apriori principle**:
 - If an itemset is frequent, then all of its subsets must also be frequent
- Apriori principle holds due to the following property of the support measure:

$$\forall X, Y : (X \subseteq Y) \Rightarrow s(X) \geq s(Y)$$

- Support of an itemset never exceeds the support of its subsets
- This is known as the **anti-monotone** property of support

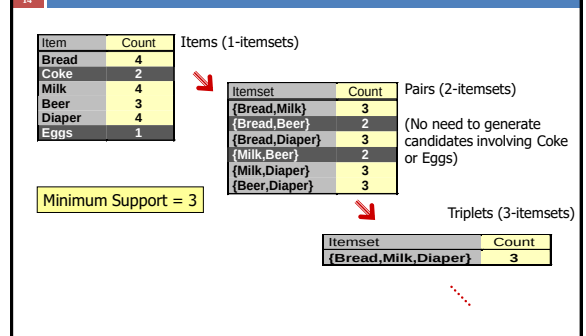
12

Illustrating Apriori Principle



13

Illustrating Apriori Principle



14

The Key Step

- Find the **frequent itemsets**: the sets of items that have minimum support
- Every subset of a frequent itemset must also be a frequent itemset, i.e., if $\{AB\}$ is a frequent itemset, both $\{A\}$ and $\{B\}$ should be a frequent itemset
- Iteratively find frequent itemsets with cardinality from 1 to k (k -itemset)
- Use the frequent itemsets to generate association rules.

15

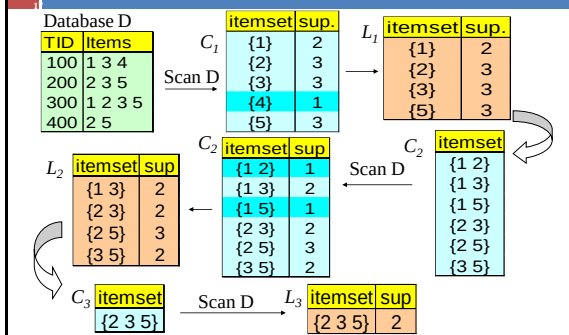
The Apriori Algorithm

C_k : Candidate itemset of size k
 L_k : frequent itemset of size k

$L_1 = \{\text{frequent items}\};$
for ($k = 1; L_k \neq \emptyset; k++$) **do begin**
 C_{k+1} = candidates generated from L_k
for each transaction t in database **do**
increment the count of all candidates in C_{k+1}
that are contained in t
 L_{k+1} = candidates in C_{k+1} with min_support
end
return $\cup_k L_k$

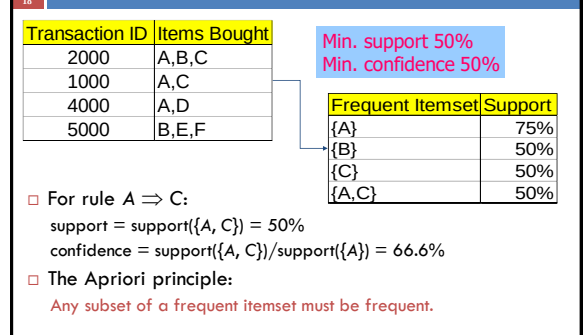
16

Apriori Algorithm – Example $\text{sup} = 2$



17

Mining Association Rules -- Example



18

Association Rules and Frequent Itemsets

19

- The Apriori algorithm calculates rules that express probabilistic relationships between items in frequent itemsets.

Ex. A rule derived from frequent itemsets containing A, B, and C might state that

IF A and B are included in a transaction,
THEN C is likely to also be included.

19

Association Rules and Frequent Itemsets

20

Antecedent and Consequent

- The IF component of an association rule is known as the antecedent.
- The THEN component is known as the consequent.
- The antecedent and the consequent are disjoint; they have no items in common.
- Oracle Data Mining supports association rules that have one or more items in the antecedent and a single item in the consequent.

20